Pre-Analysis Plan RT

Hypothesis 1

Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables

We operationalize “having daughters” using a dummy variable. We construct it to indicate per year in which the survey was run, whether the respondent reported to have had a daughter based on the child’s reported birth year (hasdaughters = 1). Any other value (not having daughters, as well as missing values or implausible data) is coded as the reference category (hasdaughters = 0). The variables we use to construct this indicator variable are kidsex* bioyear and kidgeb*.

We operationalize support for gender equality as rated support for state involvement in relieving care work duties traditionally borne by women. This includes childcare, but also care for the sick and elderly (variables plh0016 plh0017 plh0018 plh0019 plh0020 plh0021 plh0022 plh0023 plh0024 plh0025 plh0026). From these variables, we construct a latent variable support_state indicating state support via a structural equation model.

Inclusion/exclusion

We drop data from responses without associated personal or household identifiers.
We recode answers indicating different types of nonresponse or missingness in the state support variables to one category of missing.

Planned Analysis

We planned to run a linear mixed effects model with random intercepts at the individual (pid) and household (cid) levels. We predict the support for state-involvement to relieve care work (used to operationalize attitude towards gender equality measures) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year-fixed effect.

Because the model does not converge in the full data set, we change the estimation approach. We instead use a linear panel regression (GLS) with random effects at the level of participants (pid). We predict the support for state-involvement to relieve care work (used to operationalize attitude towards gender equality measures) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year-fixed effect.

Interpretation of Results

Results support Hypothesis 1 if the main effect of hasdaughters on support_state is significant at the 0.05 level and positive.

Hypothesis 2

Having daughters leads to more left-leaning political preferences.

Variables

We operationalize “having daughters” using a dummy variable. We construct it to indicate per year in which the survey was run, whether the respondent reported to have had a daughter based on the child’s reported birth year (hasdaughters = 1). Any other value (not having daughters, as well as missing values or implausible data) is coded as the reference category (hasdaughters = 0). The variables we use to construct this indicator variable are kidsex* bioyear and kidgeb*.

We operationalize left-leaning political preferences as a binary indicator variable (left_leaning). We construct this variable from reported party preference (variables plh0012_h plh0012_v6). We assess reported party preferences according to political spectrum indicated by Wahlomat 2024 for the European Elections, and set cutoffs as follows:

- left_leaning = 1 indicates a left-leaning preference, i.e. party support for SPD, Greens, Die Linke, SPD/Greens, Piraten
- left_leaning = 0 points to non-left-leaning preferences (center/right), i.e. party support for CDU, CSU, FDP, far-right, CDU/CSU, CDU/FDP, AfD

Inclusion/exclusion

We drop data from responses without associated personal or household identifiers.
We recode answers indicating different types of nonresponse or missingness in party preferences variables to one category of missing.
We drop all responses on reported party preference that indicate mixed political preferences (e.g., SPD/CDU, SPD/AfD, SPD/far-right).

Planned Analysis

We planned to run a logistic mixed effects model with random intercepts at the individual (pid) and household (cid) levels. We predict the party preference for parties on the left of the German spectrum (used to operationalize left-leaning political preferences) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year-fixed effect.

Because the model is not fully computationally reproducible, we change the estimation approach. We instead use a logistic panel regression with random effects at the level of participants (pid). We predict the party preference for parties on the left of the German spectrum (used to operationalize left-leaning political preferences) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year- fixed effect.

Interpretation of Results

Results support Hypothesis 2 if the main effect of hasdaughters on left_leaning is significant at the 0.05 level and positive.

Hypothesis 3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables

We operationalize “having daughters” using a dummy variable. We construct it to indicate per year in which the survey was run, whether the respondent reported to have had a daughter based on the child’s reported birth year (hasdaughters = 1). Any other value (not having daughters, as well as missing values or implausible data) is coded as the reference category (hasdaughters = 0). The variables we use to construct this indicator variable are kidsex* bioyear and kidgeb*.

We operationalize gender equality in the household as a binary indicator variable (household_GEM). We construct this variable from whether respondents report having the last word in financial decisions in their household. If respondents indicate they share the last word with their partner, we take this to indicate gender equality in the household (household_GEM = 1). If the respondents indicate they or their partner have the last word, this indicates the reference category (household_GEM = 0).

Inclusion/exclusion

We drop data from responses without associated personal or household identifiers.
We recode answers indicating different types of nonresponse or missingness in having the last word to one category of missing, and keep only the three categories outlined above (shared last word, partner has last word, respondent has last word).

Planned Analysis

We planned to run a logistic mixed effects model with random intercepts at the individual (pid) and household (cid) levels. We predict if the last word on financial decisions in the household (used to operationalize gender equality in the household) is shared (1 = shared, 0 = unilateral) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year-fixed effect.

Because the model is not fully computationally reproducible, we change the estimation approach. We instead use a logistic panel regression with random effects at the level of participants (pid). We predict if the last word on financial decisions in the household (used to operationalize gender equality in the household) is shared (1 = shared, 0 = unilateral) from whether the person has daughters (1 = yes, 0 = everything else), controlling for a year-fixed effect.

Interpretation of Results

Results support Hypothesis 3 if the main effect of hasdaughters on household_GEM is significant at the 0.05 level and positive.

Hypothesis 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables

We operationalize “having daughters” using a dummy variable. We construct it to indicate per year in which the survey was run, whether the respondent reported to have had a daughter based on the child’s reported birth year (hasdaughters = 1). Any other value (not having daughters, as well as missing values or implausible data) is coded as the reference category (hasdaughters = 0). The variables we use to construct this indicator variable are kidsex* bioyear and kidgeb*.

We operationalize support for gender equality as rated support for state involvement in relieving care work duties traditionally borne by women. This includes childcare, but also care for the sick and elderly (variables plh0016 plh0017 plh0018 plh0019 plh0020 plh0021 plh0022 plh0023 plh0024 plh0025 plh0026). From these variables, we construct a latent variable support_state indicating state support via a structural equation model.

We operationalize male sex as responses to variables indicating reported respondent sex (pla0009_v1 pla0009_v2 pla0009_v3) that indicate the category “male” (male = 1). As the reference category, we set responses to the category “female” (male = 0).

Inclusion/exclusion

We drop data from responses without associated personal or household identifiers.
We recode answers indicating different types of nonresponse or missingness in the state support variables to one category of missing.
We drop all responses to the variables indicating reported respondent sex that are not male or female.

Planned Analysis

We planned to run a linear mixed effects model with random intercepts at the individual (pid) and household (cid) levels. We predict the support for state-involvement to relieve care work (used to operationalize attitude towards gender equality measures) from whether the person has daughters (1 = yes, 0 = everything else) and whether they are male (1, female = 0) as well as the interaction of the two predictors, controlling for a year-fixed effect.

Because the model does not converge in the full data set, we change the estimation approach. We instead use a linear panel regression (GLS) with random effects at the level of participants (pid). We predict the support for state-involvement to relieve care work (used to operationalize attitude towards gender equality measures) from whether the person has daughters (1 = yes, 0

= everything else) as well as the interaction of the two predictors, controlling for a year-fixed effect.

Interpretation of Results

Results support Hypothesis 4 if the interaction effect of hasdaughters#male on support_state is significant at the 0.05 level and positive.